

Dharmik Umeshkumar Jha

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[Portfolio](#)

EDUCATION

Nirma University, Institute of Technology, Ahmedabad, Gujarat, India. 2023 – 2027

- **Major:** B.Tech Mechanical Engineering (7th Sem) — CGPA: 8.64/10.0
- **Minor:** Data Science
- **Relevant Coursework:** Machine Design, Finite Element Analysis, Design & Dynamics of Machines, Aerodynamics, Automation & Control, Manufacturing Processes

EXPERIENCE

Mechanical Design Intern — Shree Siddhi Vinayak Engg. Co. Summer 2026

Industrial Internship

- Contributed to the mechanical design development of an agricultural rover, creating and refining CAD models in SolidWorks. Worked on component sizing, mechanical layouts, and assembly integration based on project requirements.
- Considered manufacturability and practical design constraints during design iterations.

Society of Automotive Engineers (SAE) Nirma Collegiate Club — Team Arrow Oct 2023 – Mar 2026

Core Member (2023) → Design Lead (2025) → Captain (2025–2026)

- Led mechanical design and development activities across multiple aerospace systems, covering structural design, CAD, aerodynamic sizing, fabrication, integration, and experimental validation.
- Directed a 9-member team at SAEISS DDC 2025; led full design-build-test of a fixed-wing UAV (<4 kg empty weight, 6 kg payload capacity, and a 500 g airdrop mechanism) across a 5-month cycle.
- Led structural and mechanical development of an 800 mm-span hybrid tilt-rotor tricopter for SAE Aero Design East 2026, including propulsion sizing, mechanism integration, lightweight structural design, and autonomous transition requirements.
- Designed a sub-2 kg quadrotor airframe using Fusion 360 generative design, consolidating the frame and rotor arms into a single Nylon-12 monocoque; achieved 1.847 kg final AUW, 3.86 thrust-to-weight ratio, and 21.57 min hover endurance.
- Contributed to aerodynamic analysis, structural sizing, composite fabrication, and experimental validation across 3 national and 2 international UAV competitions.

RESEARCH PUBLICATION

- **Journal Paper:** "Design and Aerodynamic Performance Assessment of High Payload Capacity Fixed-Wing Unmanned Aerial Vehicle (UAV)."
the Journal of Flow Visualization and Image Processing, *Begell house*.
DOI: 10.1615/JFlowVisImageProc.2026064024
- **Conference Paper:** "Design Evolution and Aerodynamic Performance Assessment of Regular-Class Fixed-Wing Unmanned Aerial Vehicle (UAV)."
Presented at 12th International & 52nd National Conference on Fluid Mechanics and Fluid Power (FMFP 2025), Nirma University.

SIGNIFICANT ACADEMIC PROJECTS

Design and Development of Advanced Class UAV – SAE Aero Design East 2026 Sept 2025 – March 2026

- Designed tilt-rotor tricopter (800 mm wingspan, 1.6 kg AUW, S1223 + NACA 9325 lifting-body fuselage) capable of autonomous hover-to-cruise transition; mechanically actuated thrust-vectoring system with 110° rotation of the front thrust vectors to enable hover-to-cruise transition.
- Sized tri-motor T-Motor F80 Pro 2500KV propulsion system to deliver thrust-to-weight ratio of 2:1; static bench testing confirmed 3.6 lbs thrust at 50% throttle, providing a 36% thrust margin above hover requirement.

Aerothon 2025 – Disaster Response Quadrotor UAV

March 2025 – August 2025

- Designed a sub-2 kg quadrotor (440 mm wheelbase) for disaster response applications. Developed a single integrated airframe using **Generative Design** to optimise strength-to-weight ratio and manufacturability. Fabricated using 3D-printed PETG for lightweight structural reliability.

Design of Micro Class Aircraft – SAE Aero Design West 2025

Sept 2024 – April 2025

- Designed, fabricated, and tested a 1.6 kg micro-class UAV achieving a payload fraction >0.5 under a 450 W power limit. Achieved short takeoff (<10 ft), 5-minute endurance, and high structural integrity through aerodynamic optimisation and validation testing.

Design of Regular Class Fixed Wing Aircraft – SAE Drone Development Challenge

Oct 2023 – March 2025

- Designed, fabricated, and tested a fixed-wing UAV with empty weight <4 kg, optimised for high payload capacity within geometric constraints (wingspan <72 in; total dimensions <150 in). Constructed using balsa and aeroply with aluminium and acrylic spars for enhanced rigidity. Integrated a payload box ($10'' \times 4'' \times 4''$) with 500 g airdrop capability in conventional tail configuration.

Design of Primary Landing Gear – SAE Drone Development Challenge

Oct 2023 – March 2025

- Designed a lightweight primary landing gear for a 10 kg fixed-wing UAV using carbon-fibre rods secured with CNC-machined aluminium clamps. Integrated an expanding spring mechanism for landing impact absorption. Final assembly weighed ~ 250 g - representing less than 2.5% of total aircraft weight- while sustaining rated landing impact loads validated through drop testing.

TECHNICAL SKILLS

- **Mechanical Design & CAD:** SolidWorks, Fusion 360, Generative Design, CAD Modelling, Assembly Design
- **Structural Analysis:** ANSYS Structural, Structural Sizing, Finite Element Analysis
- **Manufacturing & Prototyping:** 3D Printing, Laser Cutting, Composite Fabrication
- **Programming & Numerical Analysis** Python (NumPy, Matplotlib), MATLAB
- **Flight Systems & Integration:** Pixhawk Autopilot, Mission Planner

HONOURS & AWARDS

- Achieved **overall 11th place** and **5th place in mission score** out of 75 international teams at SAE Aero Design West (Micro Class) 2025, Apollo XI RC Field, Van Nuys, California, USA — April 2025.
- Secured **1st position** in the Regular Class Category and **3rd position** in the Micro Class Category at the SAEISS Drone Design Challenge 2025, Karpaga Vinayaga College of Engineering & Technology, Chennai — March 2025.
- Achieved **overall 4th place** and **2nd Best Design Report** in the Regular Class Category at the SAEISS Drone Design Challenge 2024, SRM University, Chennai — September 2024.